

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger – Master NLP



ML Basics

Chapter 1

Sources: Lecture and Exercise

- Learning Paradigms, AI vs ML vs DL
 - <https://youtu.be/chfwJiXvBMA>
- Supervised Learning (Dataset + Algorithm = Predictive Model; Linear Regression Model)
 - <https://youtu.be/UQFIss3pu-k>
- Formalization Dataset (Attributes vs. Features, discrete vs continuous, classification vs regression)
 - <https://youtu.be/PK6IEeAWiRg>
- Algorithm: Model Class, Objective, Optimizer
 - <https://youtu.be/PED1OYvQUX4>
- Text Classification (Features Generation from text)
 - https://youtu.be/y4_g_Ni3bGw

Flipped Classroom Questions

- What are differences between reinforcement learning and supervised learning?
- What is a “loss function”? Name two examples.
- What is “x” und “y” in a “supervised learning dataset”?
- What is a “bag of words”-Representation?

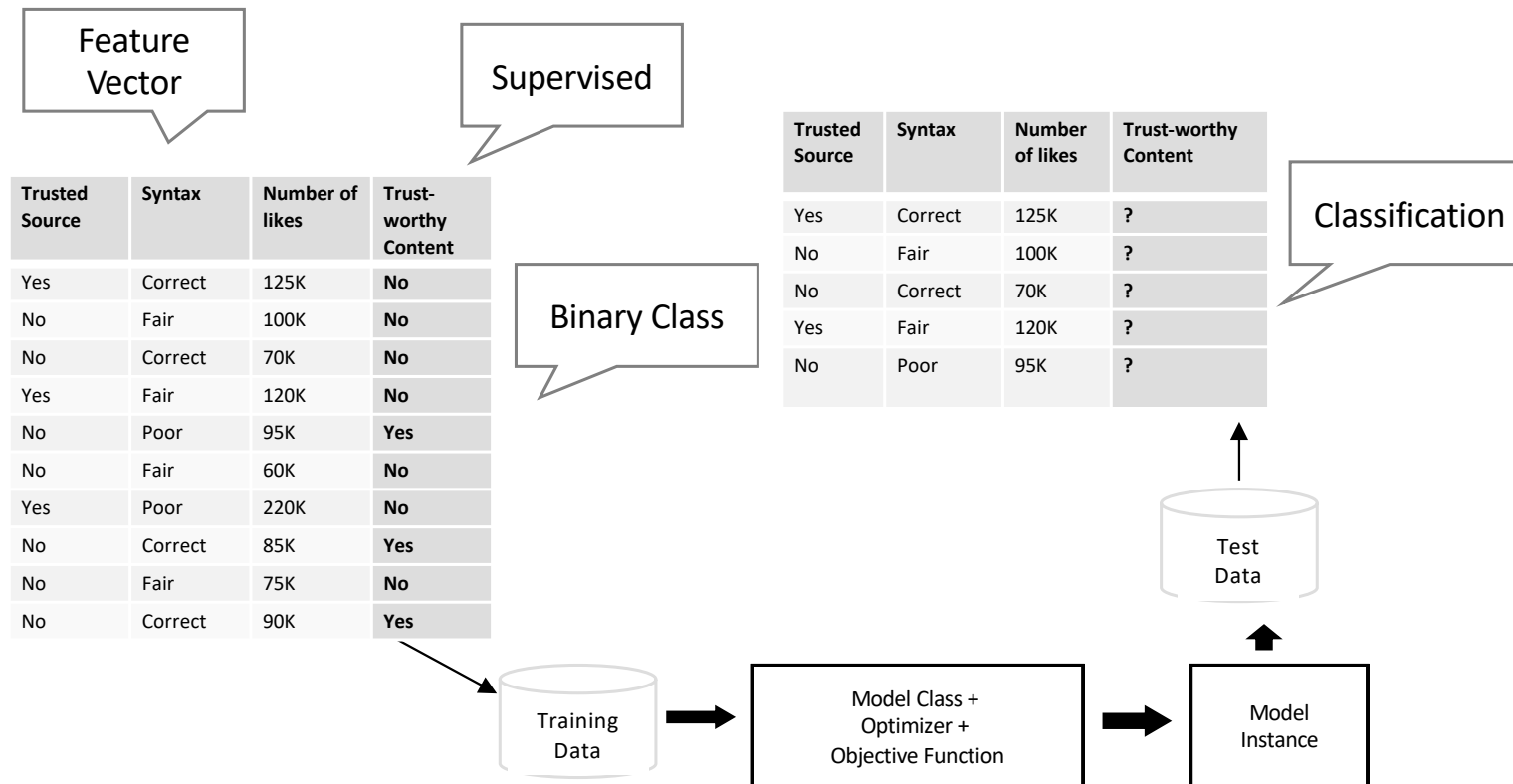
Learning Paradigms, AI vs ML vs DL

<https://youtu.be/chfwJiXvBMA>

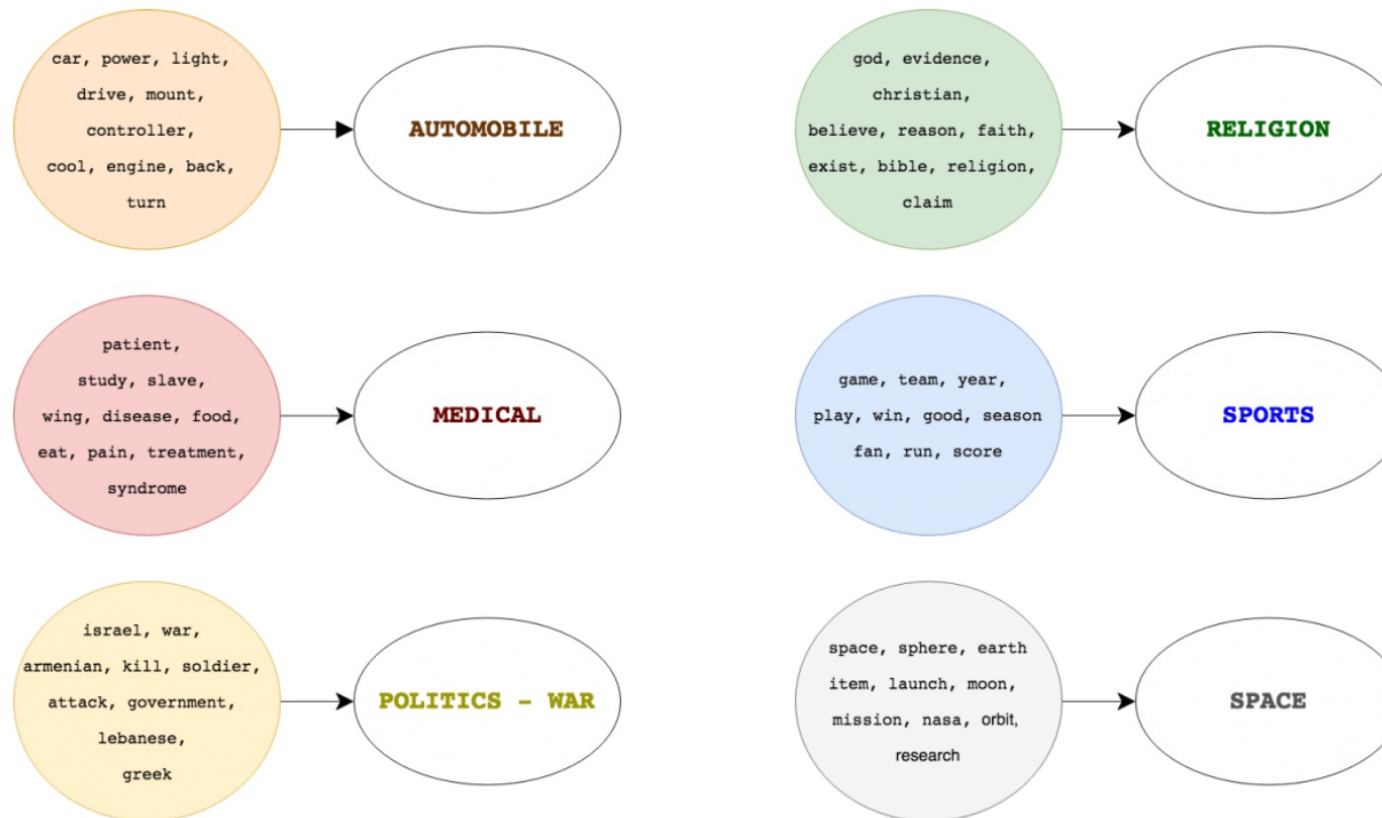
Supervised vs Unsupervised vs Reinforcement

- Supervised Learning (focus of this lecture)
 - A prediction target is given during training
 - Prediction target is often hand-labelled / manually annotated
 - Example: Spam Classification
- Unsupervised Learning (covered not specifically)
 - No specific prediction target is given
 - Let the algorithms find similarities and patterns in the data
 - Example: Find documents with similar topics
- Reinforcement Learning (covered only a little)
 - Learning to act, by (random) trial and feedback (from the environment)
 - Specify a reward that the computer receives once he reaches a certain state
 - Example: Chess / Go playing

Supervised Learning: Fake News Classification



Unsupervised Learning: Topic Modeling



<https://www.machinelearningplus.com/nlp/topic-modeling-gensim-python/#14compute-model-perplexity-and-coherence-score>

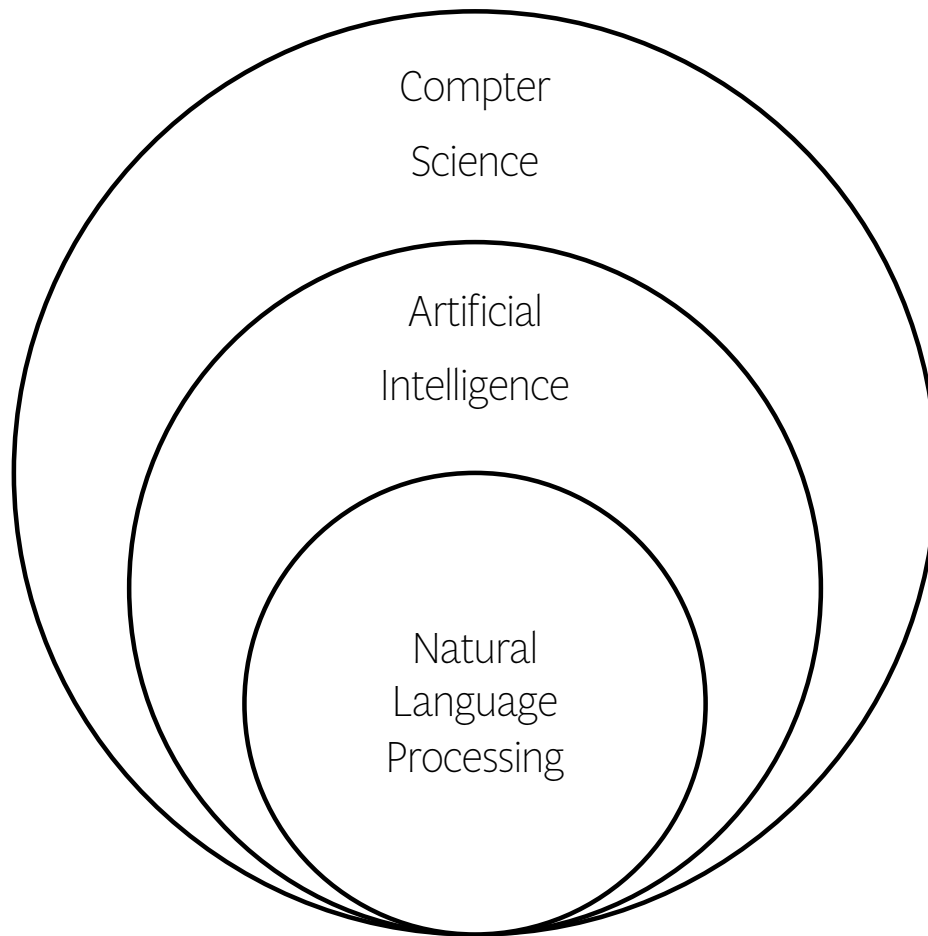
Inferring Topic from Keywords

Reinforcement Learning



<https://youtu.be/WXuK6gekUIY>

Definition and Research Area Hierarchy



- AI is an umbrella term for technologies that can **simulate** (human) intelligence.
- AI systems **mimic** cognitive capabilities, like learning from examples and solving problems.
- Natural Language Processing is a sub-area of AI

Supervised Learning: Dataset + Algorithm = Predictive Model

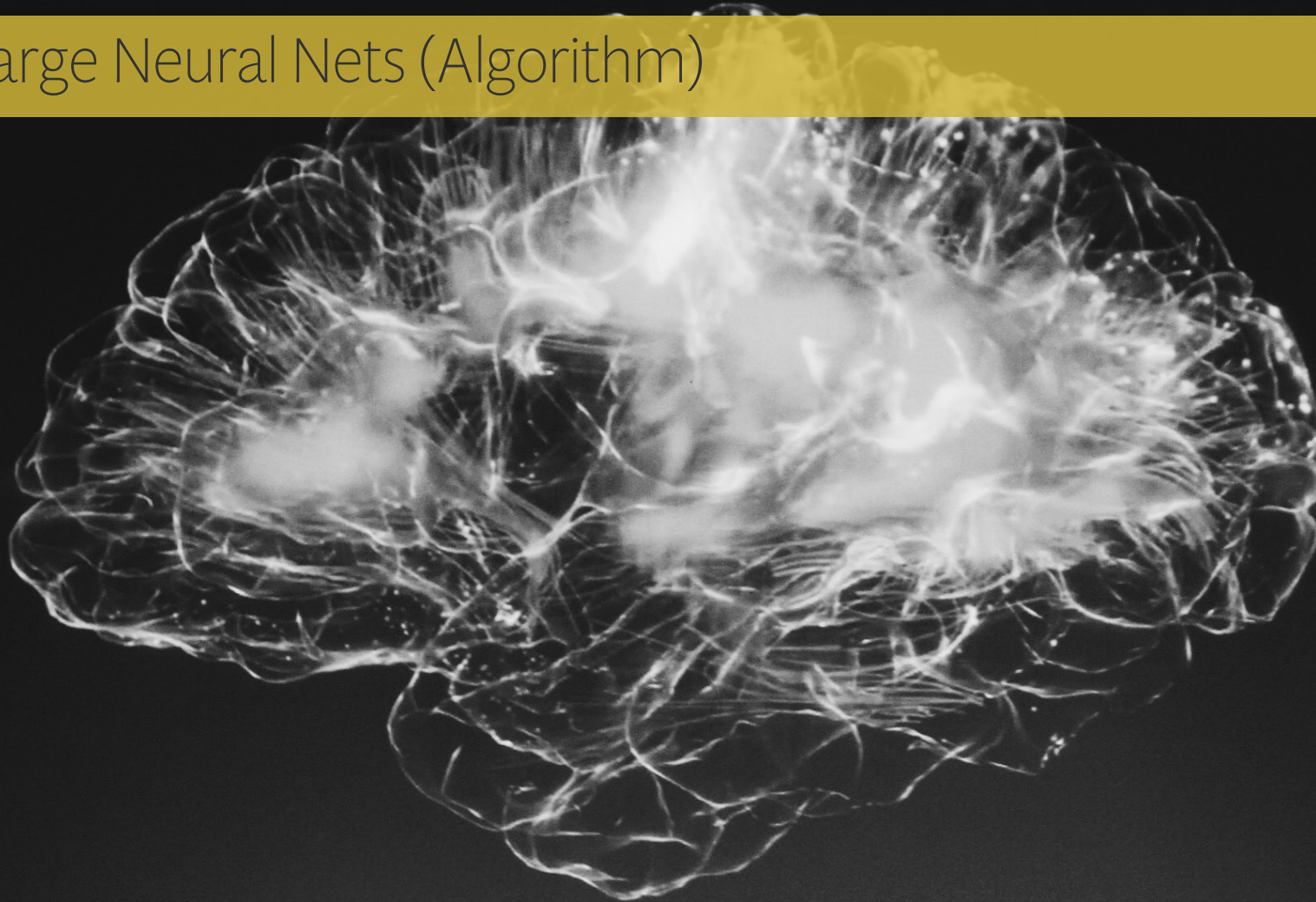
<https://youtu.be/UQFISs3pu-k>

Modern Machine Learning: Large Datasets

A photograph of a city skyline at sunset, viewed through a window. The sky is a mix of orange, yellow, and grey. Several high-rise buildings are visible, some with lights on. In the foreground, a blue neon sign is mounted on a dark surface, displaying the text "Data has a better idea" in a white, sans-serif font. The sign is illuminated with a bright blue light.

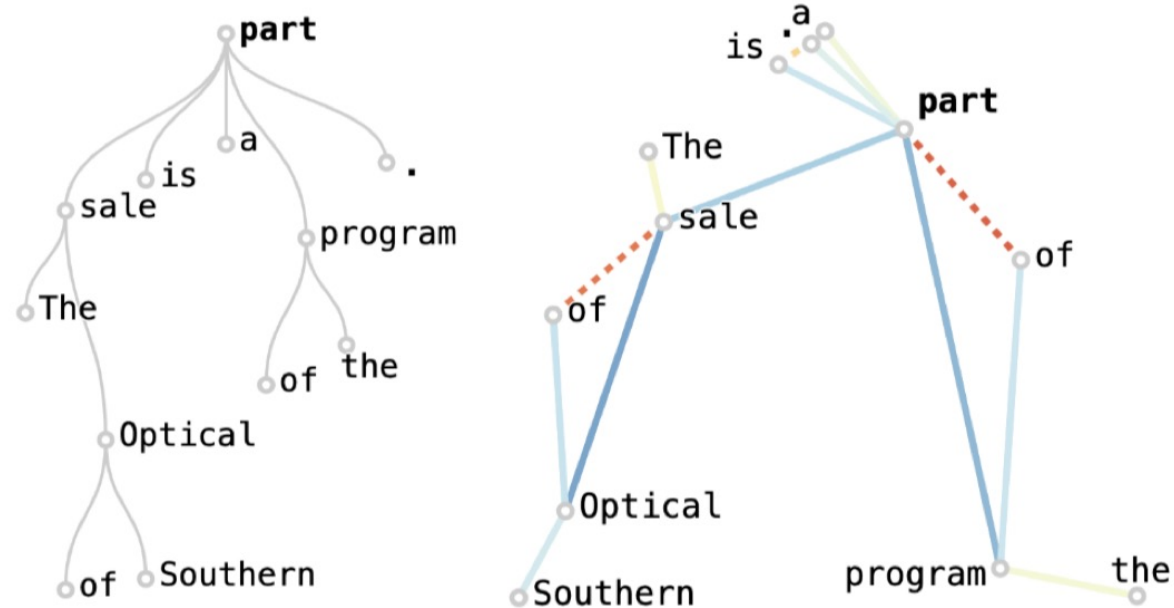
Data has a better idea

+ large Neural Nets (Algorithm)



Dataset + large Neural Net = Predictive Model (Dependency tree)

“The sale of Southern Optical is a part of the program.”



<https://arxiv.org/pdf/1906.02715v1.pdf>

Three Aspects of Machine Learning Models

Real-world Setting	Data Representation	Learning Paradigm	Task
	• Feature vector • Sequence • Graph •	• Supervised • Unsupervised • Reinforcement • ...	• Regression • Classification • Clustering • ...

Data Representation

Real-world
Setting

Data Representation	Learning Paradigm	Task
• Feature vector • Sequence • Graph •	• Supervised • Unsupervised • Reinforcement • ...	• Regression • Classification • Clustering • ...

Feature Vector			
Trusted Source	Syntax	Number of likes	Trust-worthy Content
Yes	Correct	125K	No
No	Fair	100K	No
No	Correct	70K	No

Supervised

Binary Class

Training data:

$$\text{train} := \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\} \in \mathbb{R} \times \{0, 1\}$$

Classification model: $f(x) = \hat{y}$

Misclassification rate:

$$\frac{1}{|\text{test}|} \sum_{(x,y) \in \text{test}} 1 \text{ if } y \neq \hat{y}$$

Classification by Linear Regression

- Linear Regression
$$f(\mathbf{x}_i, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j x_{i,j}$$
$$= \mathbf{x}_i^T \mathbf{w}$$
- Define $y_i = 1$ if $\mathbf{x}_i$ is assigned to class 1 and 0 otherwise
- All new x are assigned to class 1 if $f(x) > 0.5$

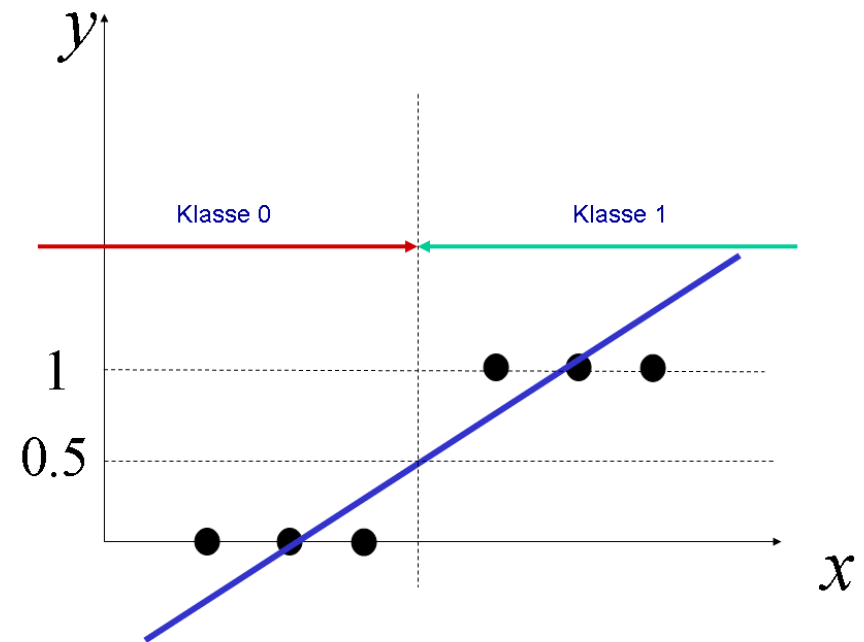
Binary Classification with Linear Regression

Optimize a linear regression model

$$Y = \langle X, w \rangle$$

for Residual Sum of Squares (RSS) loss

$$\sum_{n=1}^N (y_i - x_i^T w)^2$$



Formalization Dataset

<https://youtu.be/PK6IEeAWiRg>

Classification vs Regression, Discrete vs Continuous

- Classification:
 - Prediction of discrete feature
- Regression
 - Prediction of continuous feature

Feature Vector

Supervised

Trusted Source	Syntax	Number of likes	Trust-worthy Content
Yes	Correct	125K	No
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No	Correct	70K	No

Binary Class

Feature Vector

Supervised

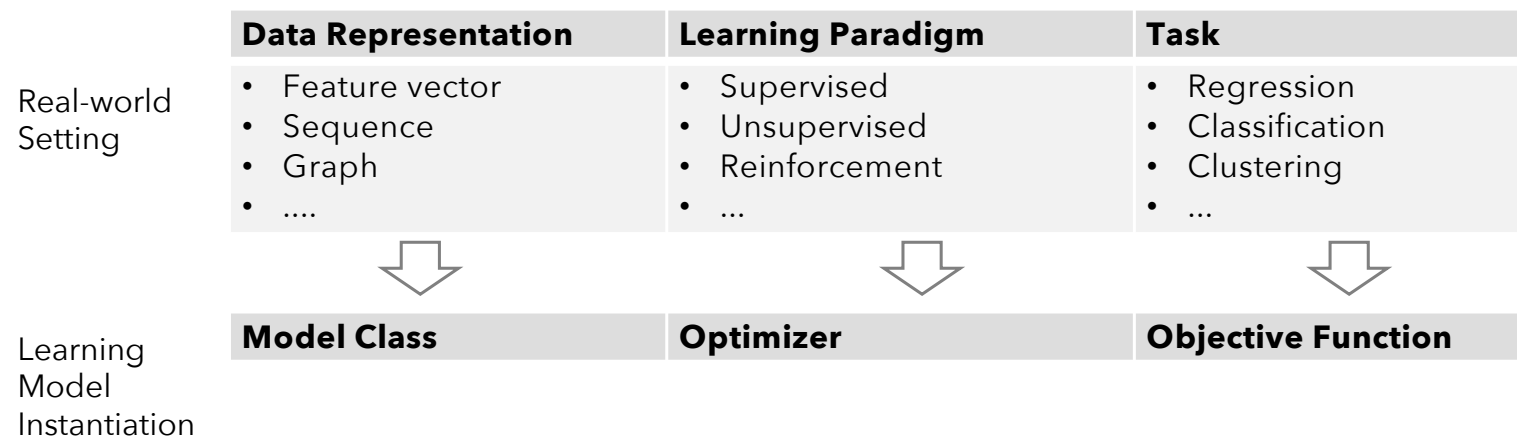
Trusted Source	Syntax	Trust-worthy Content	Number of likes
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Continuous Value

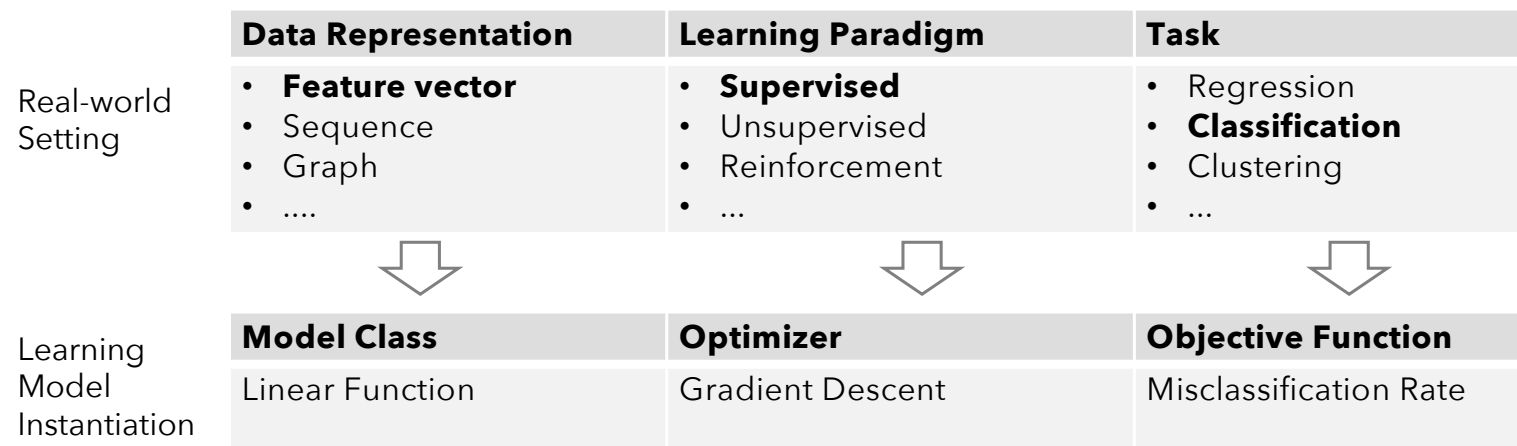
Algorithm: Model Class, Objective, Optimizer

<https://youtu.be/PED10YvQUX4>

Three Aspects of Machine Learning Models



Linear Classification



Model Notation

Real-world
Setting

Data Representation	Learning Paradigm	Task
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Feature Vector			
Trusted Source	Syntax	Number of likes	Trust-worthy Content
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Supervised

Binary Class

Training data:

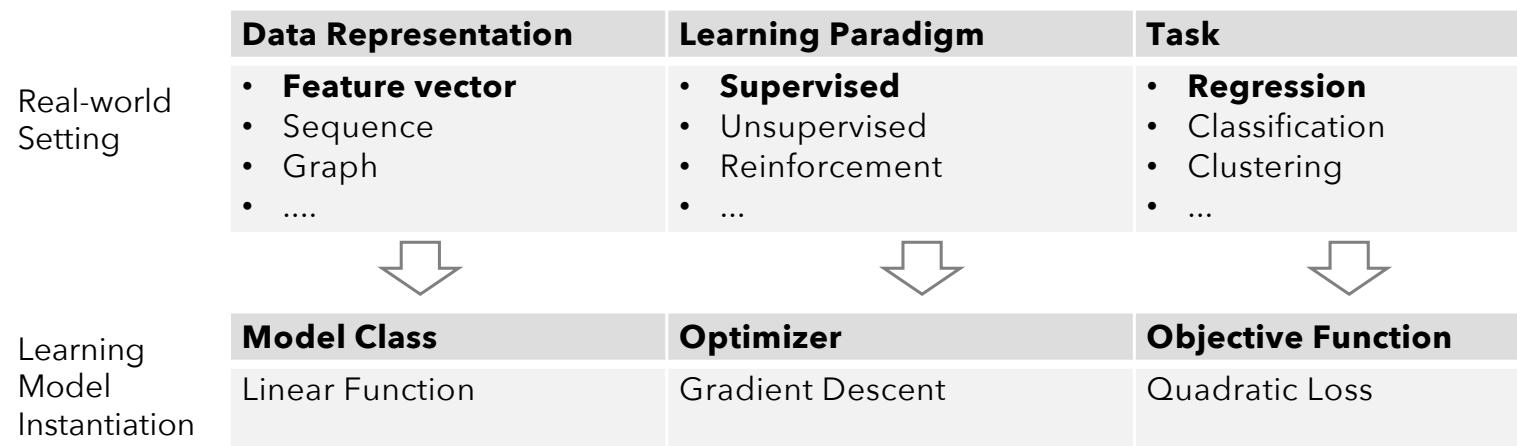
$$\text{train} := \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\} \in \mathbb{R} \times \{0, 1\}$$

Classification model: $f(x) = \hat{y}$

Objective Function: Misclassification rate:

$$\frac{1}{|\text{test}|} \sum_{(x,y) \in \text{test}} 1 \text{ if } y \neq \hat{y}$$

Classification by Linear Regression



Mean Squared Error = Quadratic Loss

- Supervised Learning:
 - Vector of inputs: $\mathbf{X} = (X_1, X_2, \dots, X_n)$
 - Single output: Y
- Define a function $f(\mathbf{x}, \mathbf{w})$
Mapping $\mathbf{x}$ to y according to parameter vector $\mathbf{w}$
- Define a loss function $\text{loss}(y, f(\mathbf{x}, \mathbf{w})) = (y - f(\mathbf{x}, \mathbf{w}))^2$
Typically, quadratic loss

Classification by Linear Regression

- Linear Regression
$$f(\mathbf{x}_i, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j x_{i,j}$$
$$= \mathbf{x}_i^T \mathbf{w}$$
- Define $y_i = 1$ if $\mathbf{x}_i$ is assigned to class 1 and 0 otherwise
- Calculate $\mathbf{w}_{LS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ as before.
- All new $\mathbf{x}$ are assigned to class 1 if $f(\mathbf{x}) > 0.5$

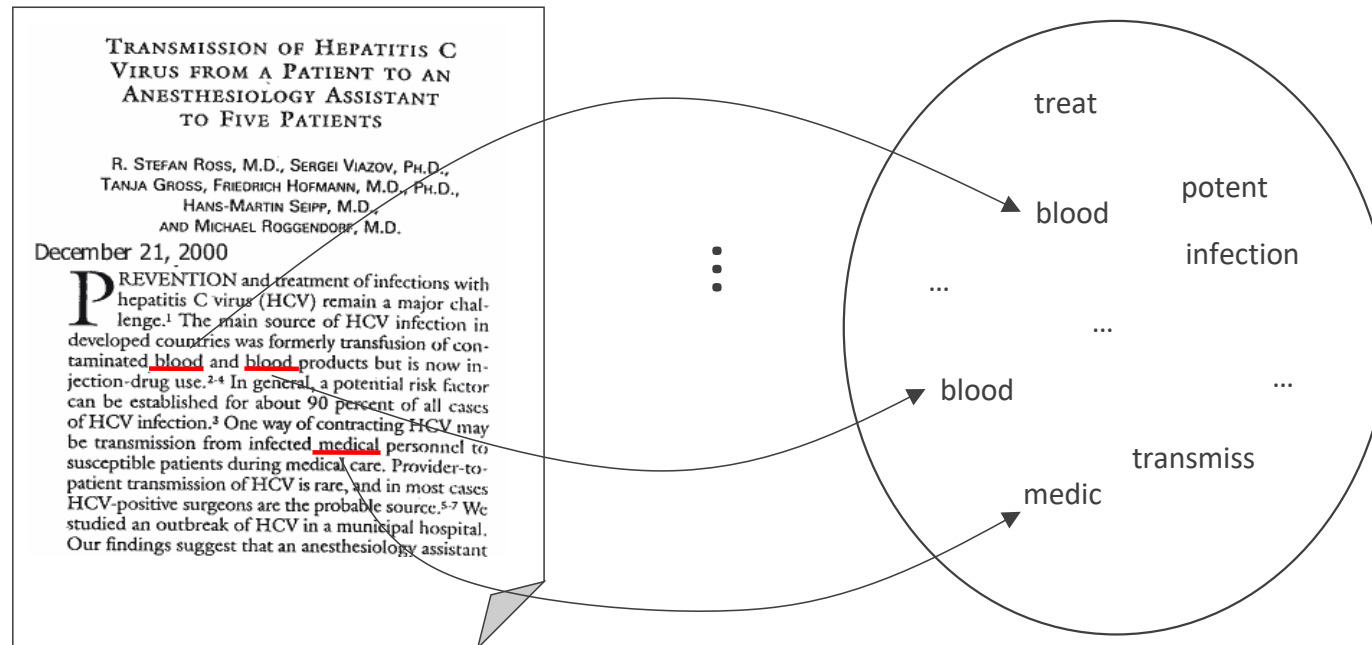
Text Classification

https://youtu.be/y4_g_Ni3bGw

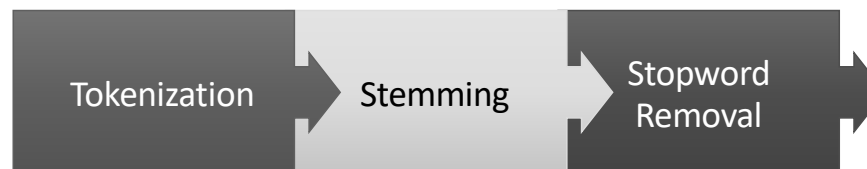
Three Aspects of Machine Learning Models

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Idea: Represent Text by a set of symbols (= words)



typical full-text indexing pipeline



Traditional NLP Pipeline Part I

1. Tokenization

- goal: segment input character sequence into "useful" tokens
- design decisions and problems:
 - set of word delimiters to use (e.g., whitespace, punctuation marks),
 - handling of special and numerical characters,
 - handling of capitalization (typically conversion to lower case),
 - handling of punctuation marks (sentence delimiter or abbreviation?);
 - different languages have different rules for compound words (e.g., "color screen" vs "Farbbildschirm,")

2. Stemming or Lemmatizing

- morphological normalization of inflected word forms to base form (e.g., "houses" → "house", "goes" → "go");
- Stemming: simple approach based on few structural rules (e.g. Porter Stemming algorithm for English language);
- Lemmatization: retrieval of base form, typically based on dictionary (can handle exceptional cases, e.g. "mice" → "mouse")

Traditional NLP Pipeline Part II

3. Stopword removal

- removal of very frequent and uninformative words,
 - typically function words as, e.g., "the", "a", "an", "of", "for",
 - e.g., SMART stopwords list for English language defines 571 stopwords (<ftp://ftp.cs.cornell.edu/pub/smart/english.stop>)
-
- Further possible (non-standard) steps:
 - separate indexing of phrases and compound words (e.g., "machine learning" ≠ "machine", "learning") - can be based on background dictionaries or statistical detection of frequent phrases (text mining again)
 - Standardization: Replace abbreviations with full expression
 - Correction of spelling
 - many more ...

Bag-of-words Representation

- Sum of the word appearances per document
- Ignore the order of words

- Example:

Vocabulary = (Hotel, Motel, is, a, this)

Hotel Hotel = [2 0 0 0 0]

this is a Hotel = [1 0 1 1 1]

this is a Motel = [0 1 1 1 1]

is this a Hotel = [1 0 1 1 1]

- Here, we use *term frequency*, i.e., number of word occurrences in a document, as the weight of each word

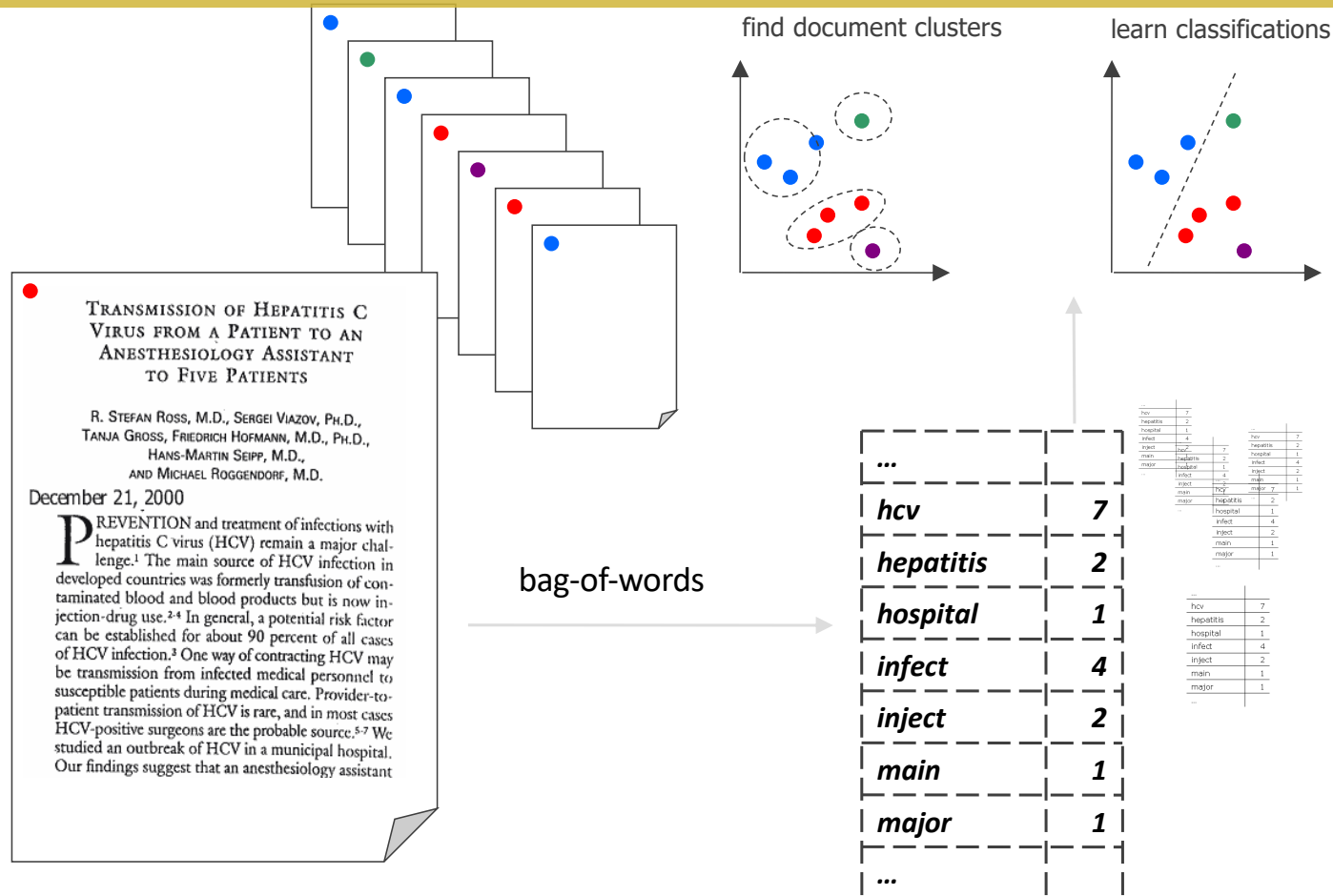
Term-Document Matrix

	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7	Document 8
Term(s) 1	10	0	1	0	0	0	0	2
Term(s) 2	0	2	0	0	0	18	0	2
Term(s) 3	0	0	0	0	0	0	0	2
Term(s) 4	6	0	0	4	6	0	0	0
Term(s) 5	0	0	0	0	0	0	0	2
Term(s) 6	0	0	1	0	0	1	0	0
Term(s) 7	0	1	8	0	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0	0

← Word Vector
(Passage Vector)

Document Vector

Text Classification and Clustering (Bag-of-Words-Model)



Text Classification

Text Classification (Text Categorization)

Text categorization (TC - a.k.a. text classification, or topic spotting), [is] the activity of labelling natural language texts with thematic categories from a predefined set [...].

*F. Sebastiani, Machine Learning in Automated Text Categorization,
ACM Computing Surveys 34(1), 2002.*

Automatic Text Classification uses:

- vector representations of text documents
- within standard classification algorithms

Text Clustering

Text Clustering

[...] the partitioning of texts into previously unseen categories [...]

Automatic Text Clustering uses

- vector representations of text documents
- within standard clustering algorithms

“The future of language is about the development of language models...”¹



1

Computerlinguistik | Universität Trier | cl.cldh.uni-trier.de